

Care Transition Efficiency & Placement Outcome Analytics

Abstract

The Unaccompanied Alien Children (UAC) Program operates as a multi-stage care transition system responsible for moving children from CBP custody to HHS care and ultimately to sponsor placement. While overall population counts are monitored regularly, process efficiency indicators are often overlooked. This project evaluates the effectiveness of the care transition pipeline using operational metrics, trend analysis, bottleneck detection, and forecasting techniques.

The analysis focuses on transfer efficiency, discharge effectiveness, backlog accumulation, and overall pipeline throughput. Interactive visualizations and a Streamlit dashboard were developed to support operational monitoring and decision-making. Results indicate a significant decline in transfer efficiency during 2025, while backlog levels remained generally controlled. Forecasting suggests that current efficiency levels may continue unless corrective measures are implemented. The project demonstrates how data analytics can support improved child welfare outcomes, faster reunification, and more effective resource allocation.

1. Introduction

The UAC Program is responsible for providing care and services to unaccompanied children after their transfer from U.S. Customs and Border Protection (CBP) to the Department of Health and Human Services (HHS). The process consists of multiple stages, including apprehension, custody, transfer, care management, and eventual reunification with a vetted sponsor.

Monitoring the efficiency of these transitions is critical because delays can increase operational costs, strain system capacity, and prolong the reunification process. Traditional reporting focuses primarily on population counts, making it difficult to identify bottlenecks and process inefficiencies. This project addresses that gap by evaluating operational performance through data-driven analytics.

2. Problem Statement

Although aggregate counts of children in custody are available, structured process-efficiency metrics are often absent. Key operational questions remain unanswered:

- How efficiently are children transferred from CBP to HHS care?
- Are discharges keeping pace with new arrivals?
- Where do care backlogs accumulate?
- Are placement outcomes improving or deteriorating over time?

Without systematic analytics, bottlenecks and inefficiencies may remain hidden, limiting opportunities for operational improvement.

3. Project Objectives

Primary Objectives

- Measure CBP to HHS transition efficiency.
- Evaluate discharge and sponsor placement outcomes.
- Identify delays and bottlenecks within the care pipeline.
- Monitor overall system throughput.

Secondary Objectives

- Support faster reunification.
- Improve operational visibility.
- Enable data-driven policy recommendations.
- Develop an interactive monitoring dashboard.

4. Dataset Description

The dataset was obtained from the U.S. Department of Health and Human Services (HHS).

The dataset contains daily operational information including:

- Date
- Children Apprehended and Placed in CBP Custody
- Children in CBP Custody
- Children Transferred out of CBP Custody
- Children in HHS Care
- Children Discharged from HHS Care

After cleaning and preprocessing, 720 valid records were available for analysis covering the period from January 2023 to December 2025.

5. Methodology

The analysis followed a structured analytics workflow:

Data Cleaning

- Removed empty records.
- Converted date fields to datetime format.
- Converted numeric columns to appropriate data types.
- Handled missing and invalid values.

KPI Engineering

The following performance metrics were developed:

Transfer Efficiency Ratio

Transfers ÷ Children in CBP Custody

Discharge Effectiveness

Discharges ÷ Children in HHS Care

Pipeline Throughput

Discharges ÷ Apprehensions

Backlog

Apprehensions – Discharges

Health Score

Composite performance indicator combining efficiency and throughput metrics.

Exploratory Data Analysis

The analysis included:

- Daily inflow versus outflow trends.
- Efficiency trend analysis.
- Monthly performance comparisons.
- Weekday pattern analysis.
- Backlog monitoring.

Forecasting

A forecasting model was developed to estimate future transfer efficiency trends and identify potential operational risks.

6. Exploratory Data Analysis

Several visualizations were developed to understand system behavior.

Daily Inflow vs Outflow

The comparison between children entering and leaving the system showed substantial fluctuations during 2023 and 2024. Activity levels decreased considerably during 2025.

Transfer Efficiency Analysis

Transfer efficiency remained relatively stable throughout 2023 and 2024, averaging between 75% and 90%. A noticeable decline occurred during 2025, suggesting potential operational changes or constraints.

Discharge Effectiveness Analysis

Discharge effectiveness remained stable during earlier periods but declined during 2025, indicating slower placement outcomes relative to the HHS care population.

Backlog Analysis

Backlog values remained predominantly negative, suggesting that outflows frequently exceeded inflows. This indicates that the system generally avoided prolonged congestion.

Weekday Analysis

Operational performance varied slightly by weekday. Thursday recorded the highest transfer efficiency, while Sunday showed the strongest discharge effectiveness performance.

7. Dashboard Development

An interactive Streamlit dashboard was developed to support operational monitoring.

The dashboard includes:

Overview

- KPI cards
- Risk status indicator

- Date range filtering
- Pipeline flow visualization

Efficiency Analytics

- Transfer efficiency trends
- Discharge effectiveness trends
- Monthly performance analysis

Bottleneck Detection

- Backlog monitoring
- Bottleneck identification
- Monthly backlog analysis

Forecasting

- 30-day transfer efficiency forecast
- Future risk assessment
- Performance outlook

Executive Recommendations

- Operational recommendations
- Policy improvement suggestions

8. Key Findings

- Transfer efficiency remained stable during 2023–2024 but declined significantly during 2025.
- Discharge effectiveness showed a similar downward trend during 2025.
- Backlog levels remained largely controlled and approached equilibrium by the end of the analysis period.
- The average health score was 16.75, indicating room for process improvement.
- Forecasting suggests that transfer efficiency may remain below historical averages if current conditions persist.

9. Recommendations

1. Monitor transfer efficiency continuously to identify emerging operational issues.
2. Investigate policy or process changes associated with the decline observed during 2025.
3. Continue maintaining low backlog levels to prevent system congestion.
4. Utilize predictive monitoring systems to identify future bottlenecks before they impact placement outcomes.
5. Expand dashboard deployment to support real-time decision-making across operational teams.

10. Conclusion

This project successfully transformed raw operational data into actionable process-efficiency insights. By evaluating transfer efficiency, discharge effectiveness, backlog accumulation, and future trends, the analysis provides a comprehensive view of care transition performance.

The results highlight the importance of continuous monitoring and predictive analytics in improving reunification timelines, reducing operational delays, and strengthening child welfare outcomes. The developed dashboard serves as a practical decision-support tool that can help stakeholders monitor system performance and respond proactively to emerging challenges.